Samar Pratap

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Education

VIT Bhopal University **Sehore, MP**
Bachelor of Technology in Computer Science and Engineering **Expected 2027**
CGPA: 8.26

Technical Skills

Languages: Python, C/C++, Java, SQL, JavaScript, HTML/CSS, Bash
Frameworks & Libraries: Pandas, NumPy, NLTK, BeautifulSoup, Streamlit, NetworkX, Pyvis, Tkinter, Node.js, Pygame-ce
Databases: MySQL
Developer Tools: Git, VS Code, Linux, Vim, Obsidian

Project Experience

Social Media Sentiment Analysis Dashboard | *Python, NLP, Pandas, NLTK, BeautifulSoup, PRAW, Streamlit*
Built an end-to-end analytics pipeline that transforms unstructured Reddit discussion data into actionable sentiment trends, mirroring a product intelligence workflow from raw data to actionable output.

- Automated data collection and preprocessing (BeautifulSoup, Reddit API) to create a reliable, repeatable data asset for ongoing trend analysis.
- Applied VADER sentiment analysis to quantify user opinion patterns and surface root-cause signals in discussion data.
- Designed an interactive Streamlit dashboard that translates raw analysis into a visual reporting tool for non-technical stakeholders.

Arcane Signatures: D&D Class Spell Analysis | *Python, Pandas, NetworkX, Pyvis*
A data analysis and visualization tool to identify unique magical signatures of D&D classes, applying pattern-discovery techniques to relational data.

- Modeled complex relational data as a network graph (NetworkX, Pandas) to identify hidden patterns and distinguishing signatures across categories.
- Built an interactive visualization layer (Pyvis, ForceAtlas2) to make abstract relational data interpretable at a glance, prioritizing clarity for a non-technical audience.

Book Vault: Library Management System | *HTML/CSS, JavaScript, Node.js, JSON*
A web-based library management system developed as a team project.

- Built backend logic using Node.js and JSON for data persistence.
- Integrated an AI chatbot and rule-based book recommendation system.

Spaceship Shooter Game | *Python, Pygame-ce*
A 2D arcade game featuring collision mechanics and scoring systems.

- Implemented pixel-perfect collision detection.
- Designed responsive restart and end-game interfaces.

Python MP3 Player | *Python, Tkinter, Pygame*
A desktop MP3 player application with playlist management.

- Developed playback controls and dynamic playlist features.
- Implemented an interactive volume control slider.